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# PRACTICAL AUDIO PROCESSING WITH MACHINE LEARNING

Understanding & Implementing

# What actually is Sound?

Imagine you clap. 🖐️

What happened?

Your hands hit each other.

That pushes air.

Air molecules move.

That movement reaches someone's ear.

Their ear vibrates.

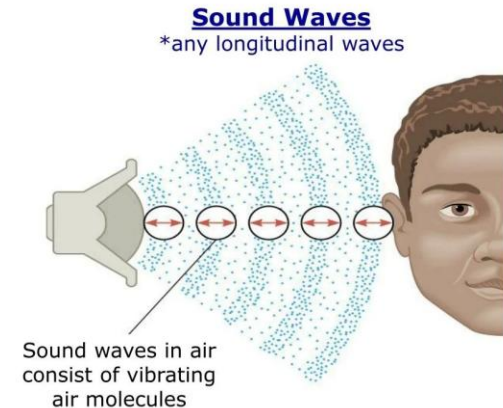
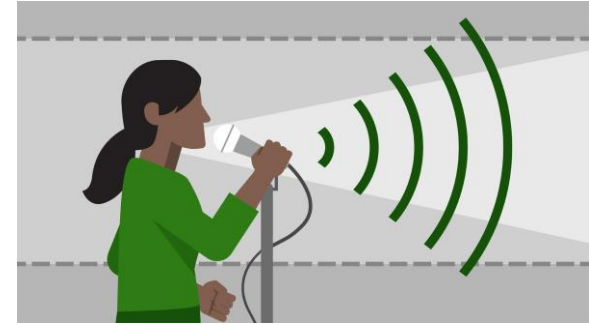
Brain interprets vibration as sound.

So...

**Sound is nothing but vibration traveling through air.**

Nothing magical.

➤ Sound needs medium to travel



# How does a microphone hear sound?

- A microphone detects changes in air pressure.
- Inside the microphone, a thin diaphragm vibrates.
- These vibrations are converted into electrical signals.
- The electrical signals are then converted into digital numbers.



Students usually think microphones understand sound.

Wrong.

Microphone understands...

Pressure.

Imagine someone says “Hello”

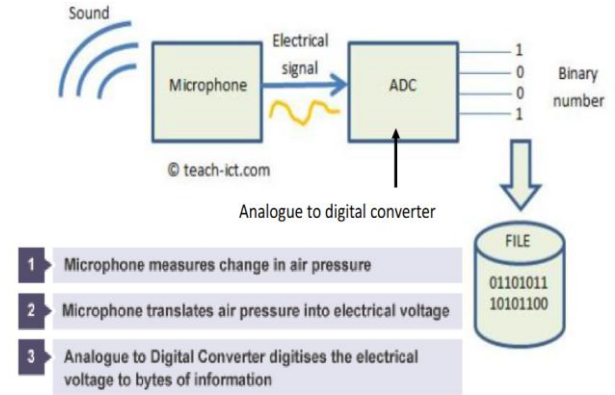
Air pressure changes like this.

High , Low, High, Low

Computers never hear sound. They only process numbers representing air pressure over time.

# How does computer store sound?

- Computers store sound as a sequence of numbers.
- Each number represents the air pressure at one moment in time.
- An audio file is simply a long list (1D array) of numerical values.
- Unlike images (2D matrices), audio is stored as one continuous sequence.



Suppose we recorded one second.

Computer stores something like[0.12,0.15,0.22,0.40,0.35,0.18,-0.10,-0.30, 0.45,...]

That's it. => Audio file

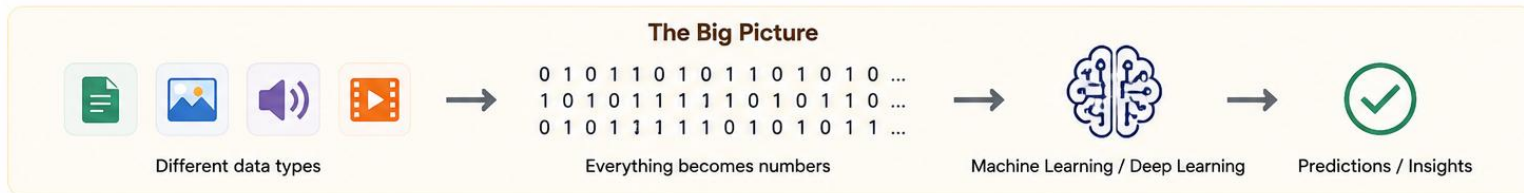
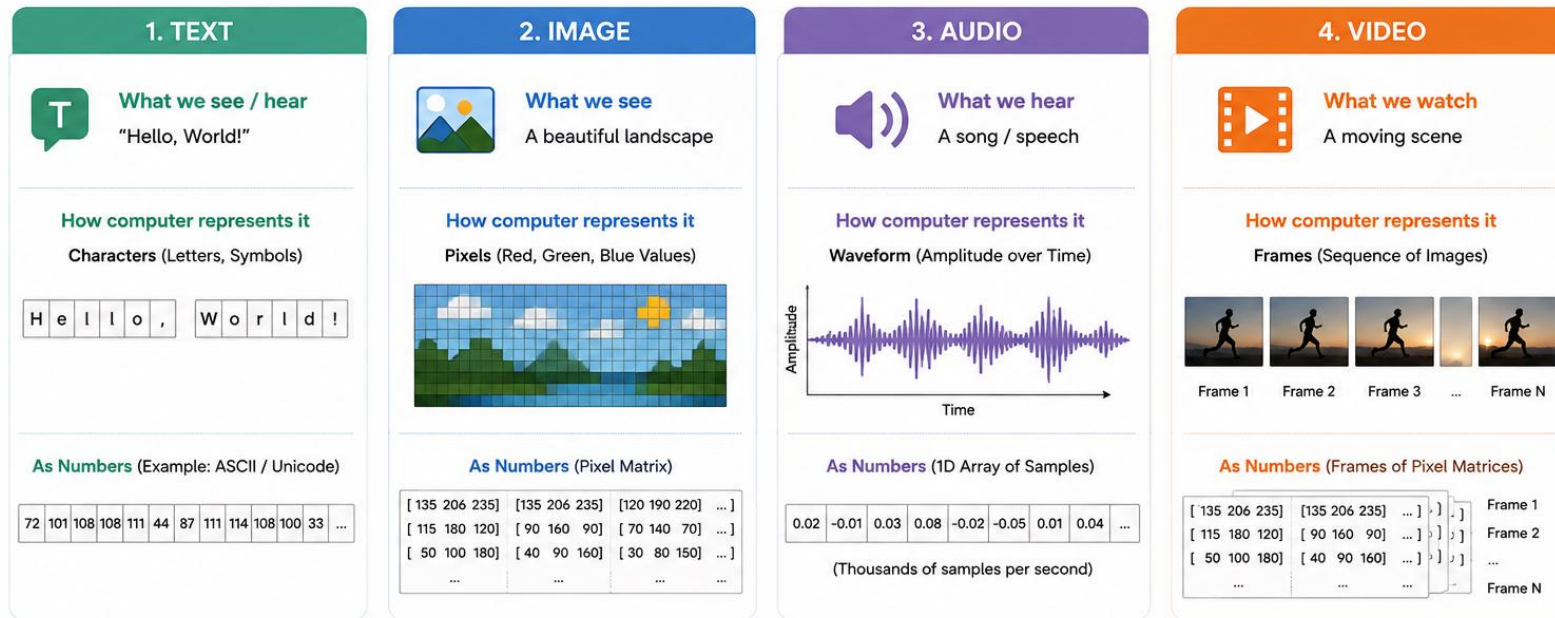
Exactly like image.

Image[[255,120,90],[130,80,50],...]

Audio[0.11,0.15,0.20,...]

# How Computers Represent Different Data Types

Computers understand everything as **numbers**.



No matter the data type – Text, Image, Audio, or Video – **computers only understand numbers.**

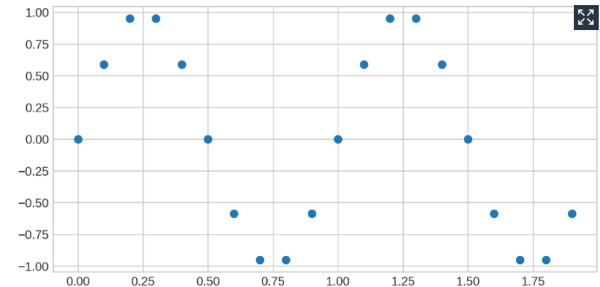
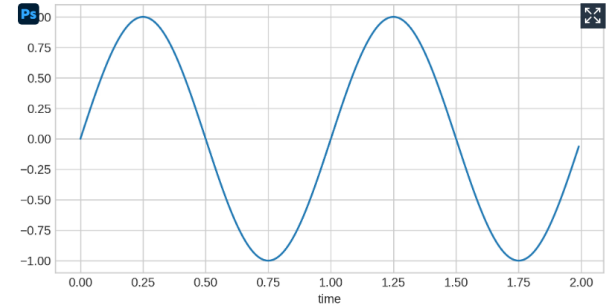
# Understanding Waveforms

## What is a Waveform?

- A waveform is the graphical representation of sound.
- One waveform can represent Speech, Music , Bird , Engine, Rain, Everything.
- It shows how air pressure changes over time.
- X-axis represents Time.
- Y-axis represents Amplitude (signal strength).

## Key Takeaway

- Waveform = Visual representation of sound.



# Sampling Rate

## How Computers Capture Sound

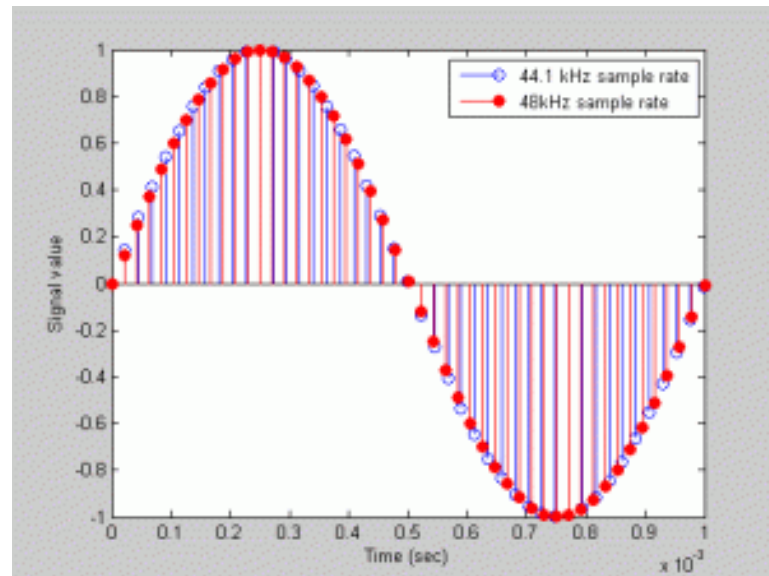
- Sound is continuous, but computers store it as discrete measurements.
- Each measurement is called a sample.
- Sampling Rate tells us how many samples are recorded every second.

## Common Sampling Rates

- 16,000 Hz
- 22,050 Hz
- 44,100 Hz (CD Quality)
- 48,000 Hz (Professional Audio)

## Why is Higher Sampling Better?

- More samples preserve more sound detail.
- Higher quality comes at the cost of larger file size.



# Amplitude vs Frequency

## Amplitude

Represents the strength of the sound wave.

Determines how loud a sound is.

Higher amplitude → Louder sound.

Lower amplitude → Softer sound.

## Frequency

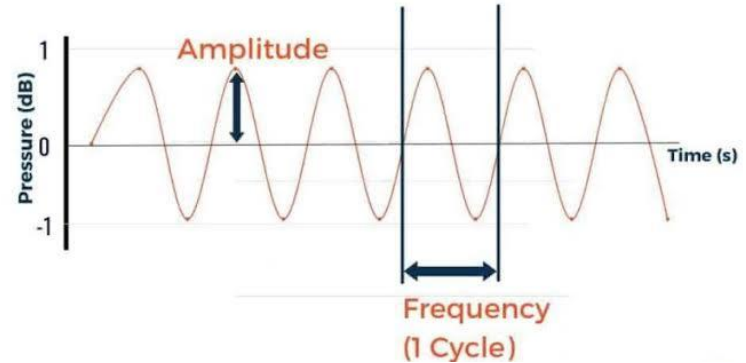
Represents how quickly a wave repeats.

Determines the pitch of the sound.

Higher frequency → High-pitched sound.

Lower frequency → Low-pitched sound.

## Amplitude & Frequency





# THE BASICS

How a computer interprets sound as data.

## CORE CONCEPTS

### WAVEFORM

A **1D array of numbers** tracking sound vibrations over time.

### SAMPLING RATE (SR)

Frequency of measurements per second.  
Standard: **22,050 Hz** or **44,100 Hz**.

### DATA SIZE

A 1-second audio file contains **22,050 data points** at standard rates.

**The Limitation:** 1D sequential audio is mathematically expensive and hard for basic algorithms to process directly over long durations.

## Why Don't We Use Raw Audio Directly?

### Challenge

Raw audio contains thousands of samples every second.

Example:

1 second  $\approx$  44,100 values

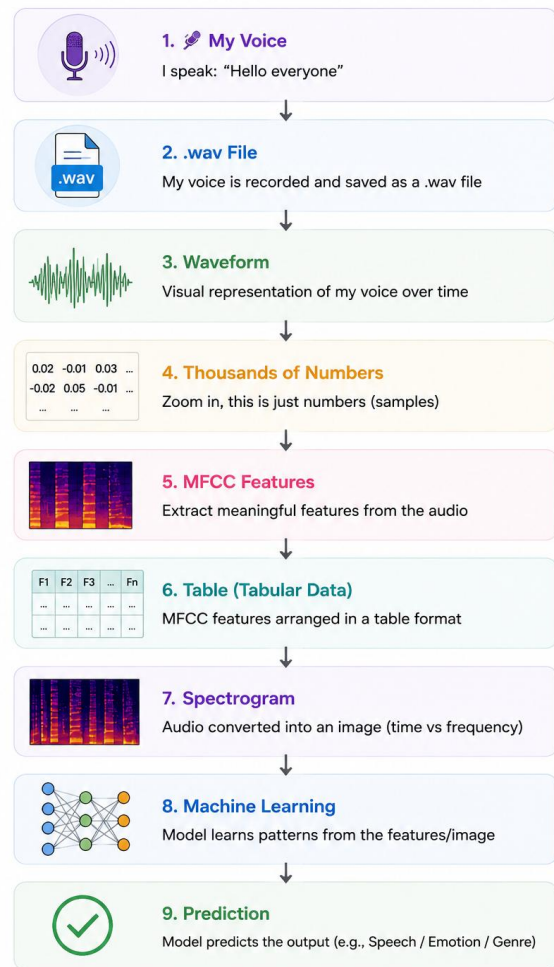
10 seconds  $\approx$  441,000 values

Processing such large sequences is computationally expensive.

**Solution:** Instead of using raw audio, we extract meaningful features.  
Feature extraction reduces complexity while preserving important information.

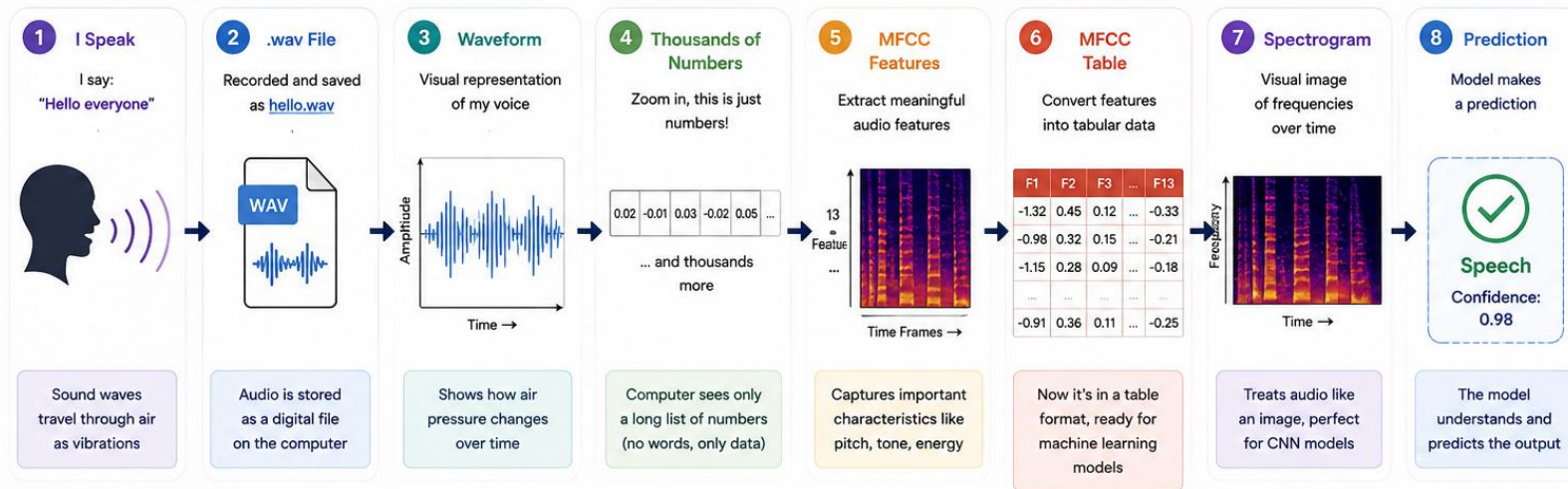


# LIVE DEMO

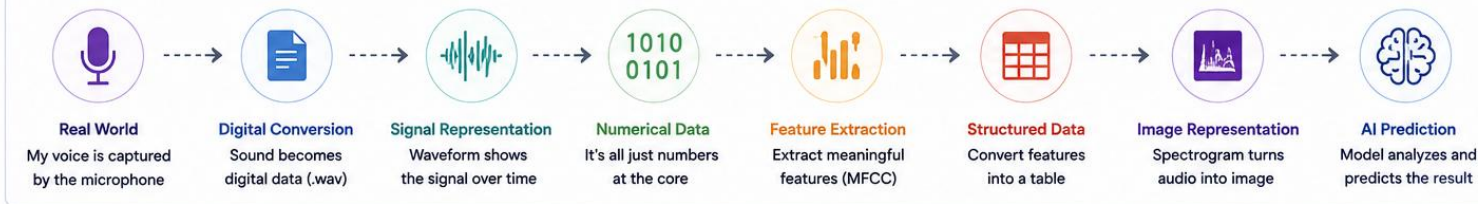


# Live Journey of My Voice

From the microphone to intelligent prediction



## What Just Happened?



**This is the complete journey of any audio in Machine Learning!**

From sound waves → numbers → features → models → intelligent predictions.

# THE FRAMEWORK

Turning sound into actionable data.



## APPROACH 1: TABULAR FEATURES

**MFCCs (Mel-Frequency Cepstral Coefficients):** Features that mimic how human ears perceive pitch and tone. Treated as structured data for classical ML like Random Forests.



## APPROACH 2: TRANSFER LEARNING

Leveraging massive pre-trained transformer checkpoints like **Wav2Vec2** or **Hubert** for direct high-level feature extraction.



## APPROACH 3: AUDIO AS GRAPHICS

Transforming sounds into 2D images (Spectrograms) and deploying **Computer Vision architectures** like Custom 2D-CNNs.

# MFCC (Mel Frequency Cepstral Coefficients)

## ➤ What is MFCC?

MFCC is one of the most popular audio features.

It captures characteristics that resemble how humans perceive sound.

Instead of thousands of samples, it produces a compact representation.

Commonly used in speech recognition and audio classification.

## ➤ Why Use MFCC?

Smaller data size

Faster training

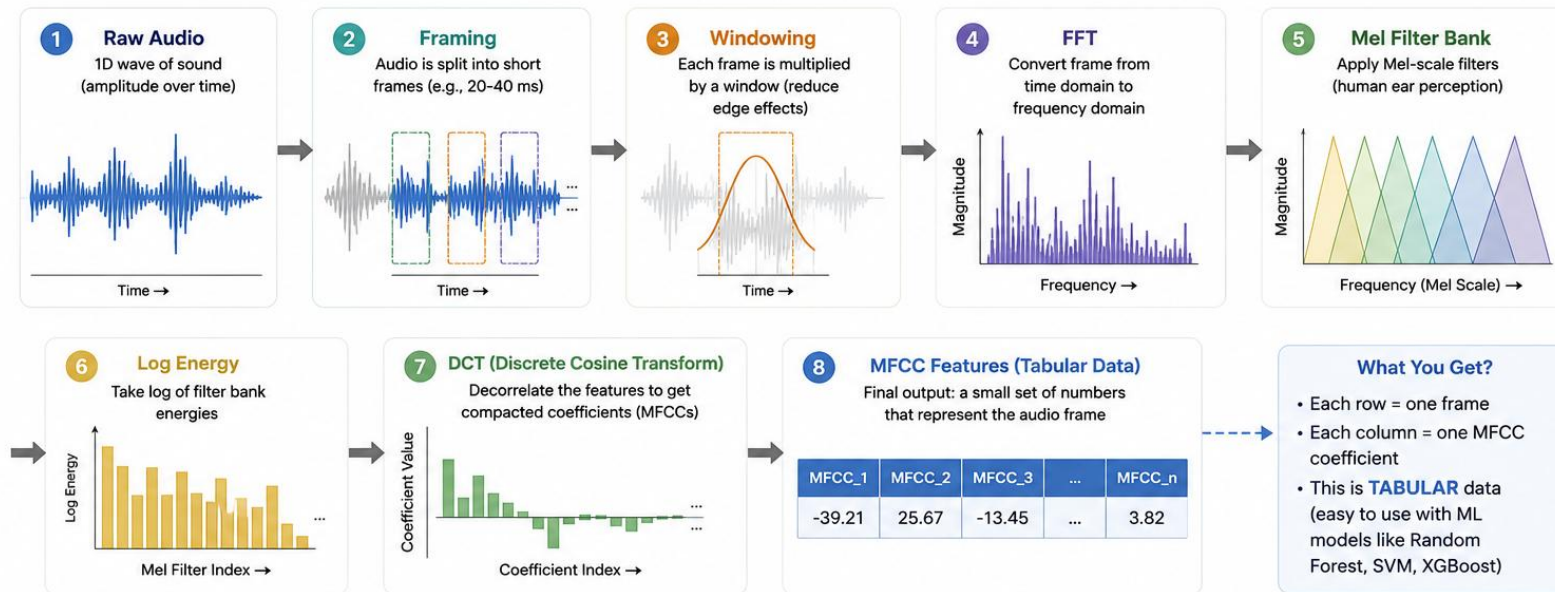
Better performance with classical ML models

## ➤ Common Models 1. Random Forest 2. SVMXGBoost

MFCC converts raw audio into meaningful numerical features.

# MFCC (Mel Frequency Cepstral Coefficients)

Converting Raw Audio into Meaningful Features (Tabular Data)



## Why MFCC Works?

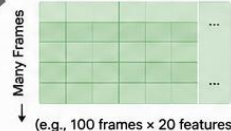
- Mimics how human ears perceive sound (Mel scale)
- Reduces thousands of samples into ~13-40 features
- Captures important characteristics like pitch, tone, and timbre
- Makes machine learning faster and more effective

## Example: 1 Second Audio

Raw Samples  
(44,100 values)

[0.02, -0.01, 0.03, ...]

MFCC Features  
(20 coefficients per frame)



Tabular Data  
(Ready for ML Models)

| MFCC_1 | MFCC_2 | ... | MFCC_20 |
|--------|--------|-----|---------|
| -39.21 | 25.67  | ... | 3.82    |
| -37.11 | 21.43  | ... | 2.11    |
| ...    | ...    | ... | ...     |
| -40.02 | 26.10  | ... | 3.01    |

## Use with ML Models



MFCC transforms raw, complex audio waves into simple, compact numerical features that machines can easily learn from.

# Spectrogram

## ➤ Converting Sound into Images

A spectrogram is a visual representation of sound.

It shows:

- Time on the X-axis
- Frequency on the Y-axis

Color/Brightness represents intensity

## ➤ Why Use Spectrograms?

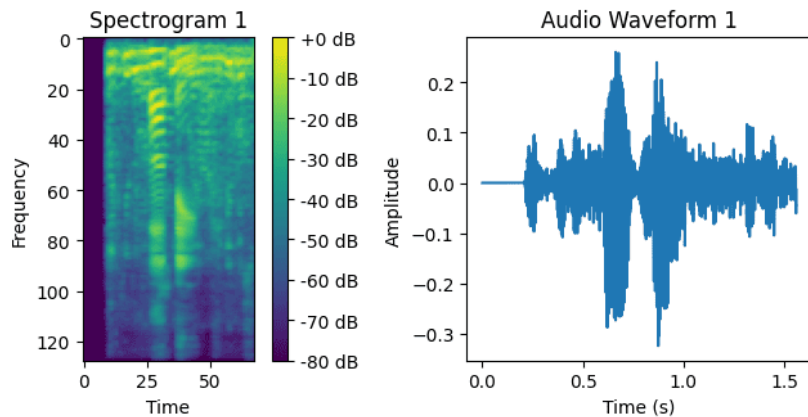
Converts audio into images.

Enables the use of Computer Vision models.

CNNs can recognize patterns in spectrograms.

## ➤ Spectrogram = Image representation of sound.

Spectrogram Comparison for Class: puppy





# SOUND TO VISION

The bridge between audio and computer vision.

## THE MEL-SPECTROGRAM

A visual map showing how **frequencies change over time**. Maps Time (X-axis) against Frequency (Y-axis), with brightness representing loudness (dB).

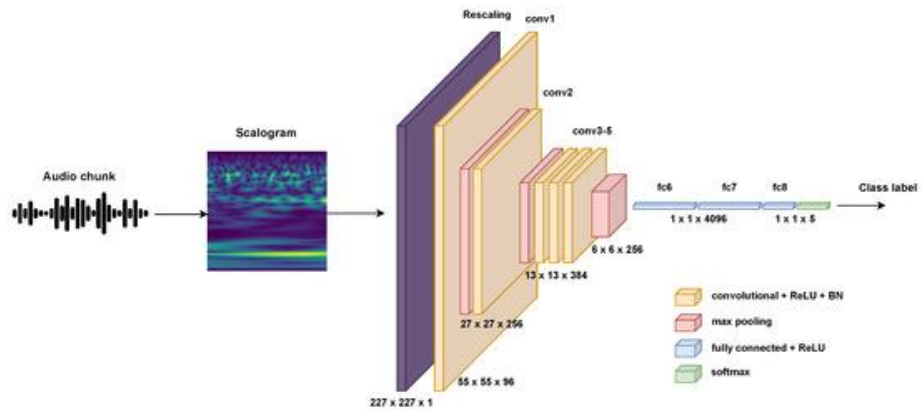
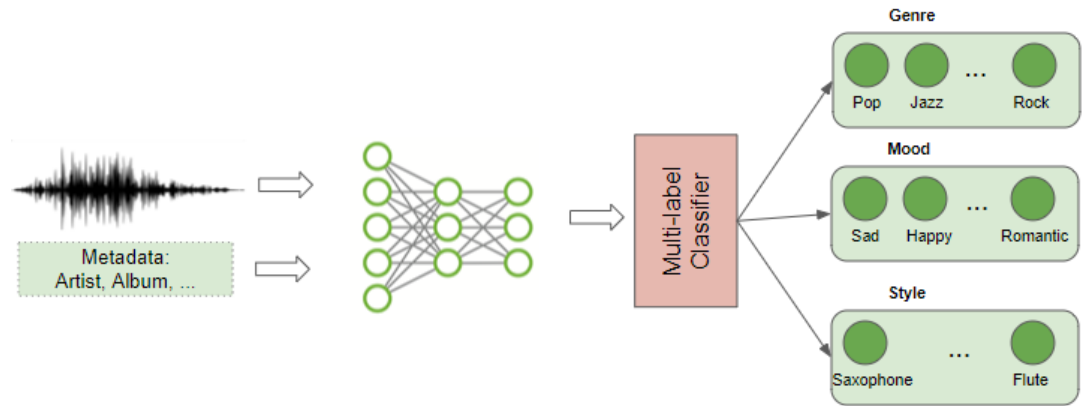
## THE COMPUTER VISION LINK

Spectrograms allow standard **CNN models** to process audio exactly like an image, leveraging spatial pattern recognition.

## DATA REGULARIZATION

Trimming and padding varying audio lengths into a standard matrix (e.g., **[64, 32]**) for secure neural network input.

**1D WAVEFORM → 2D SPECTROGRAM IMAGE**



# BEHIND THE SCENES

How the neural network decodes visual sound.

## 01 CONVOLUTIONAL LAYER 1

**16 Filters** scanning the spectrogram to capture basic textures, edge detection, and fundamental pitch transitions.

## 02 MAXPOOLING

Reducing spatial dimensions to focus on **high-energy frequency areas**, ensuring translation invariance.

## 03 CONVOLUTIONAL LAYER 2

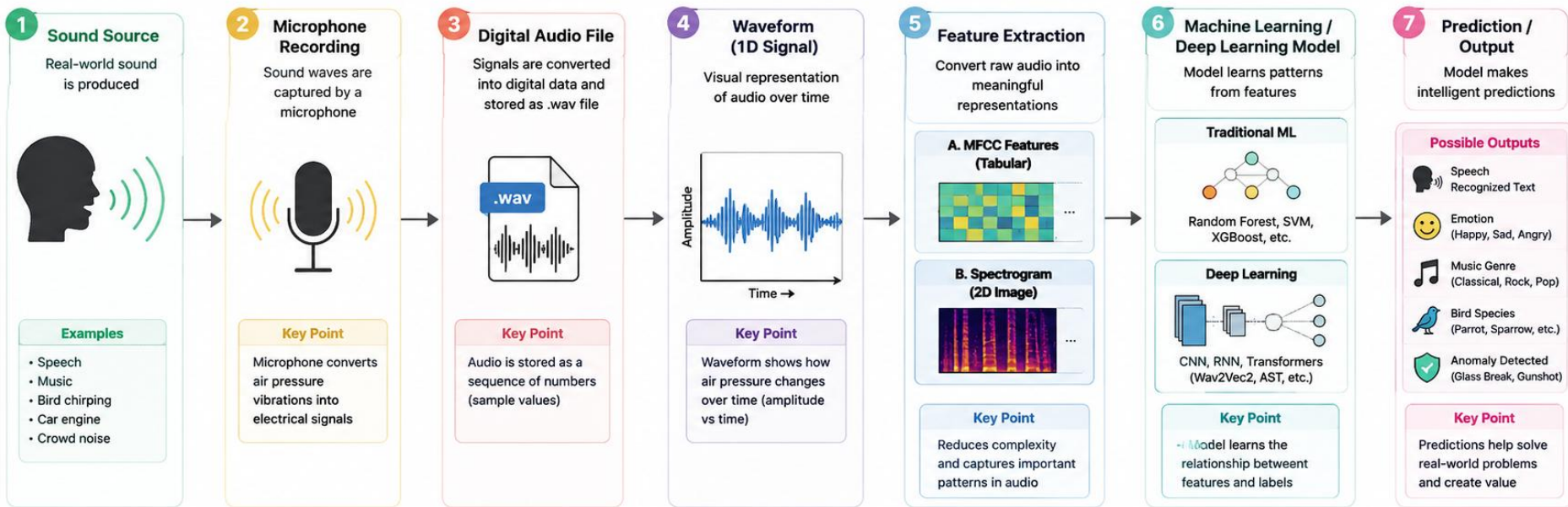
**32 Filters** mapping complex, combined vocal signatures and temporal patterns across the frequency spectrum.

## 04 FULLY CONNECTED CLASSIFIER

Flattening vectors into a dense layer to predict **discrete target categories** with a Softmax output.

# Complete Audio Machine Learning Pipeline

From Sound to Prediction



## Real-World Applications



**Speech Recognition**  
Voice assistants, transcription



**Emotion Detection**  
Call centers, mental health analysis



**Music Classification**  
Recommendation systems, music organization



**Environmental Monitoring**  
Bird sound detection, noise pollution monitoring



**Healthcare**  
Heart sound analysis, cough detection, sleep monitoring



**Security & Surveillance**  
Gunshot detection, vehicle monitoring, anomaly detection

# TOOLKIT & STRATEGY

The essential tools and the path to implementation.

## PYTHON TOOLKIT

### Librosa

The industry standard to **load audio** and extract features like MFCCs.

### Soundfile

A reliable tool to **save and write** audio files back to storage.

### Matplotlib

Used to **visually display** waveforms and spectrograms.

## EXECUTION PLAN

### 01. Preparation

Use **librosa.load()** to read files into numerical arrays.

### 02. Feature Extraction

Convert files into **MFCCs** for tabular datasets.

### 03. Modeling

Start with **Random Forest/SVM**, then move to CNNs.

## DEPLOYMENT

Moving beyond the notebook:  
Transitioning from Google Colab to  
a fully functioning local framework.

## THE WEB APPLICATION

### Production Stack



**Streamlit:** Rapid UI development for ML tools.



**PyTorch:** High-performance tensor computation.



**Soundfile:** Robust audio I/O handling.

### Architecture Benefits



Allows non-technical users to easily **drag and drop** standard .wav files.



Bypasses complex OS driver dependencies by loading files via **safe numerical data arrays**.

# GLOBAL IMPACT

Audio ML is solving critical challenges across industries, from digital safety to environmental preservation.



## SECURITY AI

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Detecting synthetic voice cloning and DeepFake media scams to protect digital identity and financial assets.



## URBAN INTEL

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Identifying vehicle crashes, glass shattering, and traffic anomalies via smart audio sensors for safer cities.



## ECO-AI

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Monitoring wildlife patterns and flagging illegal logging activities automatically in remote, protected areas.

### Example Applications

-  Speech Recognition
-  Emotion Detection
-  Music Genre Classification
-  Bird Sound Detection
-  Medical Diagnosis
-  Deepfake Voice Detection

## NEXT STEPS

The journey into Audio AI starts with high-quality data and the right tools.

### DATA REPOSITORIES

- **Kaggle Audio**

Balanced datasets for emotion and demographic profiling.

- **OpenSLR**

Massive library for multi-lingual speech training resources.

- **Mozilla Common Voice**

Clean, crowdsourced public voice datasets for global use.

### FRAMEWORKS

- **torchaudio**

PyTorch's native library for audio signal processing.

- **librosa**

The industry standard for music and audio analysis in Python.

- **Streamlit**

The fastest way to build and share data apps for deployment.



# THANK YOU.

## QUESTIONS & DISCUSSION

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